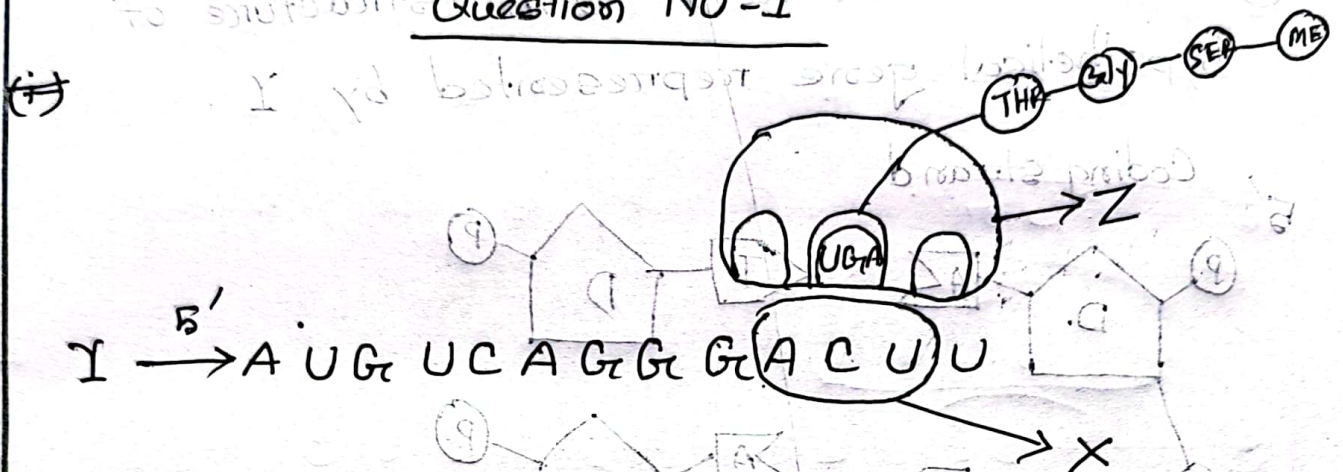


Tutorial - 1

Question NO - 1



(i) The process depicted in the diagram is called translation.

(ii) The component 'X' is called smaller subunit of ribosome and its function is decoding the genetic information.

(iii) The component 'Z' is called larger subunit of ribosome.

The large subunit of a ribosome, in both prokaryotes and eukaryotes, plays a crucial role in protein synthesis. It contains the peptidyl transferase center, which is responsible for catalyzing the formation of peptide bonds.

(iv) It is called mRNA.

(vi) Illustrate the process identified in (i) for the first two codons.

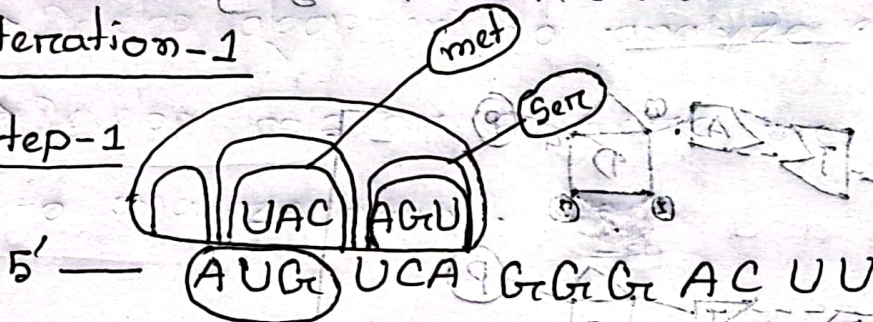
Initiation:

5' — AUG —



Iteration-1

Step-1



Step-2



Step-3



two codon translation
process.

Answer to the Q. NO-2

DNA → TAC GCT TCG AAG CTC AAC CGG CACT
mRNA → AUG CGA AGC UUC GAG UUG CCG UGA
 met Arg Ser Phe Glu Leu Pro Stop
 ↓
 start

if the nitrogenous base position 5 is being mutated to nitrogenous base A, then the mRNA & protein sequence will change

DNA → TAC GCA TCG AAG CTC AAC CGG CACT
mRNA → AUG CGU AGC UUC GAG UUG CCG UGA
 met Arg Ser Phe Glu Leu Pro Stop
 ↓
 start

Here we see that if the 5th nitrogenous base mutated to A then the codon in that position is changed from CGA to CGU.

and CGA & CGU both represent the same amino acid. So, There shouldn't be any problem.

③ Bioinformatics is an interdisciplinary field that combines or merge biology, computer science, statistics/math to solve biological problem.

④ Goals of bioinformatics.

→ Construct & merge biological database.

→ Develop algorithm to analyze data.

→ Develop tools or software or webbrowser

to analyze data.